Stomatal traits across cytotypes — short summary

Preliminary results from automated analysis of 3,253 micrographs

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Summary

We automatically detected and measured stomata in **3,253 micrographs** from **540 plants** (5 populations; four cytotypes; both leaf sides), giving **105,354 detected stomata**. Key results:

- **Cell size follows ploidy, but not linearly.** Synthetic neotetraploids (neo_4x) have the largest stomata (23.9 µm abaxial, +27% vs. nat_2x), while established natural tetraploids (nat_4x) are intermediate (20.6 µm, +10%). Diploids ory_2x and nat_2x are similar (18.5 and 18.7 µm).
- **Density compensates for size in neopolyploids.** Abaxial density drops from 254 mm^{-2} in nat_2x to 164 mm^{-2} in neo_4x (-35%), whereas nat_4x (237 mm^{-2}, -7%) does not differ significantly from nat_2x. ory_2x has the highest density (297 mm^{-2}).
- **Theoretical conductance.** Estimated $g_{s\,\max}$ is lowest in neo_4x (0.76 mol m^{-2} s^{-1} abaxial), while nat_4x (0.92) matches nat_2x (0.91); ory_2x is highest (1.03). This is consistent with natural tetraploids having re-balanced size and density after polyploidisation, whereas neopolyploids have not.
- **Leaf sides.** All cytotypes are amphistomatous, with consistently higher density on the abaxial surface.

Methods in brief

Image analysis. A YOLO11 instance-segmentation model (classes: whole stoma, pore) was retrained with additional hard-negative background images and run tile-wise on full-resolution micrographs. For each stoma we measured length, width, guard-cell width and pore aperture; size statistics use only well-focused, non-edge stomata. Pore aperture is refined at sub-pixel resolution from intensity profiles across the pore. Density = all detected stomata / imaged leaf area.

Theoretical conductance follows Franks & Farquhar (2001): $g_{s\,\max} = \frac{d}{v}\,\frac{D\,a_{\max}}{l+\frac{\pi}{2}\sqrt{a_{\max}/\pi}}$, with density D, pore area $a_{\max}$ (from pore length and aperture) and pore depth l approximated by guard-cell width.

Statistics. Each trait was analysed per micrograph with a linear mixed model, `trait ~ cytotype × leaf side + (1 | population) + (1 | population:plant)`, accounting for repeated images per plant. Means are model-estimated marginal means (EMMs); pairwise contrasts are Tukey-adjusted within leaf side.

Table 1: Model-estimated marginal means per cytotype and leaf side.

	Abaxial			Adaxial		
Cytotype	Length (µm)	Density (mm^{-2})	gsmax	Length (µm)	Density (mm^{-2})	gsmax
ory_2x	18.5	296.9	1.03	19.0	243.0	0.84
nat_2x	18.7	254.4	0.91	19.1	208.3	0.74
nat_4x	20.6	236.9	0.92	21.4	211.7	0.84
neo_4x	23.9	164.2	0.76	24.6	138.0	0.64

gsmax in mol m^{-2} s^{-1}.

Table 2: Selected pairwise differences (row minus reference), Tukey-adjusted within leaf side. *** p < 0.001, ** p < 0.01, * p < 0.05, ns = not significant.

	Abaxial			Adaxial		
Contrast	Length	Density	gsmax	Length	Density	gsmax
nat_4x - nat_2x	+1.9 ***	-17.4 ns	+0.00 ns	+2.3 ***	+3.5 ns	+0.10 ns
neo_4x - nat_2x	+5.1 ***	-90.2 ***	-0.15 ***	+5.5 ***	-70.3 ***	-0.09 *
neo_4x - nat_4x	+3.2 ***	-72.8 ***	-0.15 **	+3.2 ***	-73.8 ***	-0.20 ***
ory_2x - nat_2x	-0.2 ns	+42.5 ***	+0.12 **	-0.1 ns	+34.8 **	+0.11 **

Results

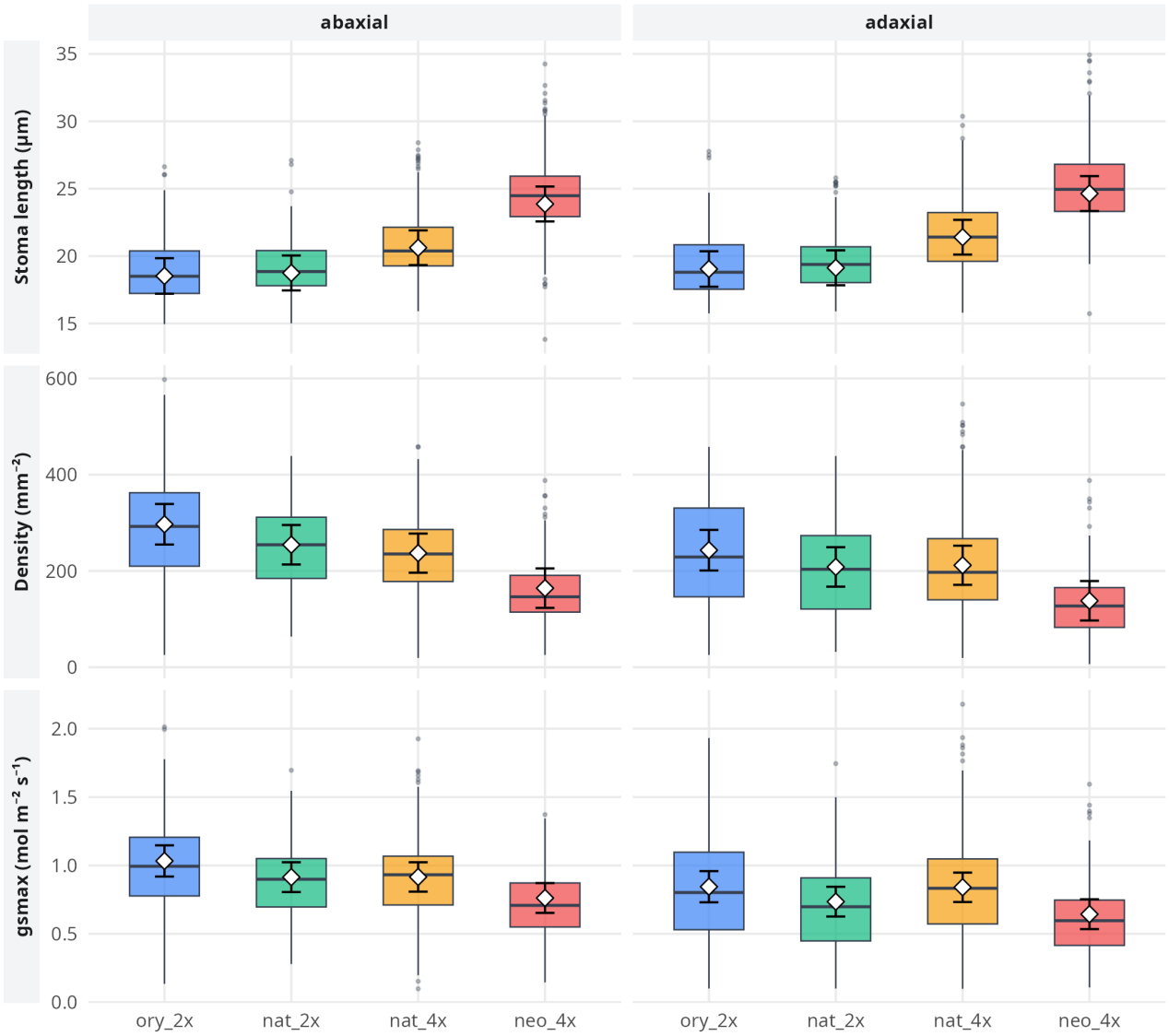


Figure 1: Stoma length, stomatal density and theoretical maximum conductance per micrograph. Boxes show image-level distributions; white diamonds with bars show model EMMs ± 95% CI.

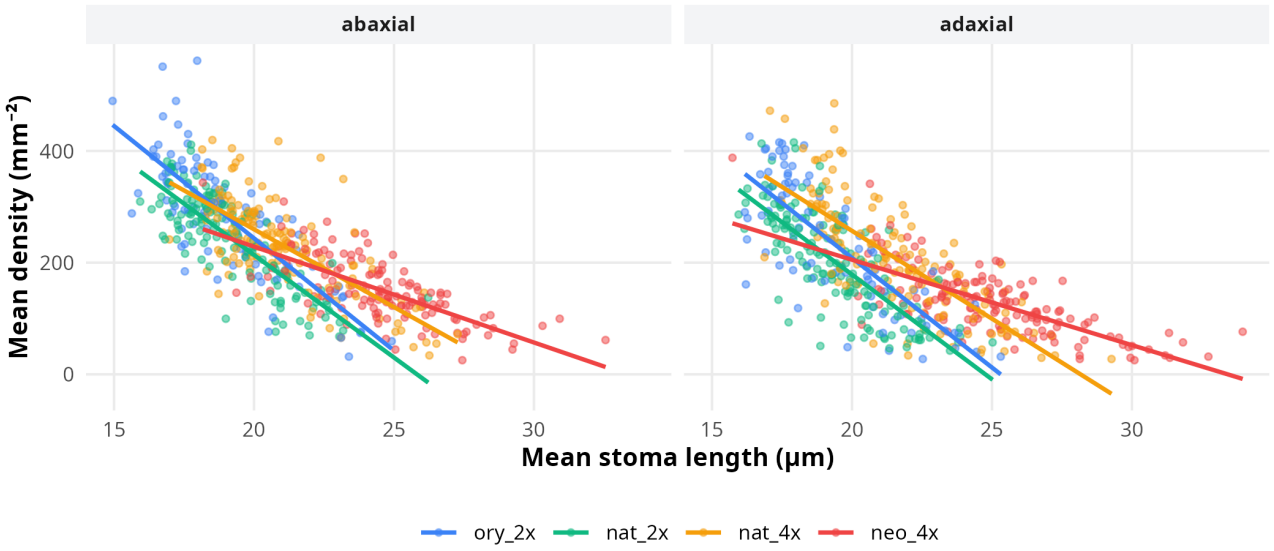


Figure 2: Size–density trade-off. Each point is a plant mean; lines are per-cytotype linear fits.

Caveats

- Results are **preliminary**; measurements come from an automated pipeline and have not yet been fully validated against manual counts on an independent set.
- $g_{s\,\max}$ is a *theoretical* anatomical estimate using guard-cell width as a proxy for pore depth and the imaged (not maximal) aperture; it is best used for relative comparisons among cytotypes, not as absolute conductance.
- With only 5 populations, the population-level variance component is estimated imprecisely.

Full analysis with code, ANOVA tables and additional traits: `docs/analysis.html` (project website).